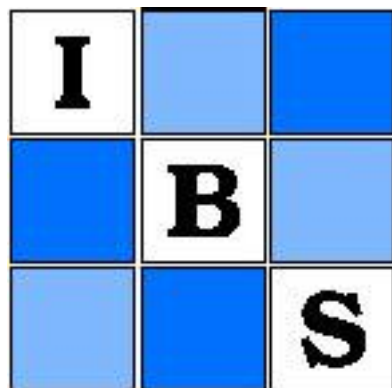




The Contributions of Jerome Cornfield to the Theory of Statistics

Author(s): Marvin Zelen

Source: *Biometrics*, Mar., 1982, Vol. 38, Proceedings of "Current Topics in Biostatistics and Epidemiology." A Memorial Symposium in Honor of Jerome Cornfield (Mar., 1982), pp. 11-15

Published by: International Biometric Society

Stable URL: <https://www.jstor.org/stable/2529849>

REFERENCES

Linked references are available on JSTOR for this article:

https://www.jstor.org/stable/2529849?seq=1&cid=pdf-reference#references_tab_contents

You may need to log in to JSTOR to access the linked references.

JSTOR is a not-for-profit service that helps scholars, researchers, and students discover, use, and build upon a wide range of content in a trusted digital archive. We use information technology and tools to increase productivity and facilitate new forms of scholarship. For more information about JSTOR, please contact support@jstor.org.

Your use of the JSTOR archive indicates your acceptance of the Terms & Conditions of Use, available at <https://about.jstor.org/terms>



International Biometric Society is collaborating with JSTOR to digitize, preserve and extend access to *Biometrics*

JSTOR

The Contributions of Jerome Cornfield to the Theory of Statistics

Marvin Zelen

Harvard School of Public Health and
Sidney Farber Cancer Institute, 677 Huntington Avenue
Boston, Massachusetts 02115, U.S.A.

SUMMARY

This paper is a review of the contributions of Jerome Cornfield to the theory of statistics. It discusses several highlights of his theoretical work as well as describing his philosophy relating theory to application. The three areas discussed are: linear programming, urn sampling and its generalizations to the analysis of variance, and Bayesian inference. It is not widely known that Jerome Cornfield was perhaps the first to formulate and approximately solve the linear programming problem in 1941. His formulation was made for the famous “Diet Problem”. An early publication introduced the method of indicator random variables in the context of urn sampling. This simple method allowed straight-forward calculations of the low order moments for estimates arising from sampling finite populations and was later generalized to the two-way analysis of variance. The application of the urn sampling model to the analysis of variance served to illuminate how one chooses proper error terms for making tests in the analysis of variance table. Jerome Cornfield’s philosophy on applications of statistics was dominated by a Bayesian outlook. His theoretical contributions in the past two decades were mainly concerned with the development of Bayesian ideas and methods. A brief survey is made of his main contributions to this area. A particularly noteworthy result was his demonstration that for the two-sample slippage problem of location, the likelihood function under a permutation setting is uninformative for the slippage parameter. However, the posterior distribution differs from the prior distribution despite the fact that the likelihood is uninformative.

1. Introduction

It is a great personal honor to be invited to discuss the contributions of Jerome Cornfield to the theory of statistics. In discussing his work it seems more natural to refer to him as Jerry as this was how he was known to all his colleagues. I shall do so in this paper.

This article will not be a complete review of Jerry’s theoretical contributions to statistics. Instead several highlights will be discussed in an attempt to impart the breadth, spirit, and insight of his contributions. The three areas which will be discussed are linear programming, urn sampling models, and Bayesian inference.

Before discussing Jerry’s work it is important to know something about his philosophy on the theory and practice of statistics. In his 1974 Presidential Address to the American Statistical Association he stated, “The usual dichotomy between theory and practice is a false one, detrimental to both theory and practice. Critical attention to application exposes shortcomings and limitations in existing theory which, when corrected, leads to new theory, new applications and a new orthodoxy in a loop which is never ending and where convergence properties are in any event unknown” (Cornfield, 1975).

Jerry’s theoretical contributions were nearly always motivated from applications in which

Key words : Analysis of variance; Bayesian analysis; Linear programming; Urn sampling.

he was personally involved. They were often deep and original contributions not only to the theory of statistics, but also to the subject matter which motivated his investigations.

2. Linear Programming and the Diet Problem

It is not well known, but Jerry was one of the first investigators to formulate and approximately solve the linear programming problem. Jerry’s first position was at the Bureau of Labor Statistics. At the Bureau, in 1941, he formulated a problem which has become widely known as the “Diet Problem”. It is a famous problem in the literature of linear programming because it was the first economic problem ever solved by the explicit use of this method.

The basic problem is that an acceptable diet must meet certain qualitative and quantitative characteristics. For example, there must be a minimum number of calories, a minimum amount of each of several vitamins, etc. Also, each food item has a specified cost per unit amount. The quality of the diet can be specified by the mathematical sum of the component parts. The optimization problem is to choose which foods should be included so that the diet is acceptable and has minimal total costs. This is the classic linear programming problem of minimizing a linear function subject to given inequality restraints.

The authoritative book on linear programming by Dorfman, Samuelson, and Solow (1958) credits Jerry with being the first person to formulate the linear programming problem and find an approximate solution. His work appeared in 1941 in an unpublished Bureau of Labor Statistics memorandum. The diet problem was eventually solved by the economist Stigler (1945), not using linear programming-type methods, and by Dantzig and Laderman in 1947 using the general linear programming formulation. Today, linear programming is regarded as one of the most important developments made in economic theory since the end of World War II.

3. Theory of Sampling, Urn Sampling Models and the Analysis of Variance

In 1944, Jerry published a short paper entitled, “On sampling from finite populations”. This paper introduced the idea of an indicator random variable as a way to obtain moments of the sample average in sampling from populations having a finite number of elements. The basic idea is both simple and ingenious. It can be stated as follows:

Let $X_1, X_2, \dots, X_N$ be a population of fixed elements from which a sample of size n are to be drawn at random without replacement. Define for $i = 1, 2, \dots, N$ the quantities $\{a_i\}$ where

$$a_i = \begin{cases} 1 & \text{if the } i\text{th element is in sample} \\ 0 & \text{otherwise.} \end{cases}$$

Then the sample mean can be written

$$\bar{X}' = \sum_{i=1}^N a_i x_i / N.$$

The random variables in the above expression are the $\{a_i\}$ whereas the $\{X_i\}$ are fixed elements.

This simple formulation of the use of indicator random variables greatly reduces the tedium of obtaining low order moments for the sample mean. The obvious extensions to more complicated survey sampling schemes have made it relatively easy to obtain the low order moments associated with the necessary estimators. We now refer to this formulation as

an *urn sampling model*. The model assumes an urn having N balls such that the value of the i th ball is X_i . The sampling scheme consists of drawing n balls at random without replacement.

The idea of indicator random variables is now routinely used in nearly all formulations of problems involving finite populations. This method is especially important in obtaining low order moments of test statistics used in the theory of distribution-free procedures.

Early in 1953, I received a manuscript to referee from the *Annals of Mathematical Statistics*. The paper was authored by Jerry. It showed how urn sampling models could be used as a model for two-way analysis of variance. At this time, there was much misunderstanding about how to choose appropriate error terms to test the various hypotheses summarized by the calculations in the analysis of variance table. The summary paper by Eisenhart (1947) on models in the analysis of variance and the investigations by Pitman (1938) and Welch (1937) on randomization were not rich enough to encompass the varied applications encountered by practitioners of the analysis of variance.

Coincidentally, I had been reading an unpublished paper by John Tukey which was written in 1949 on this same problem. The results in Jerry's paper and in John Tukey's unpublished paper were closely related. My referee's report recommended that they consider a joint publication. Both Jerry and John Tukey agreed and their joint paper appeared three years later (Cornfield and Tukey, 1956). Independently in the early 1950's Wilk and Kempthorne (1955) had also been working on this same problem. The techniques used by both Cornfield-Tukey and Wilk-Kempthorne were based on the urn sampling model of Jerry's 1944 paper.

The extension of the urn model to the two-way analysis of variance was to consider a two-way classification of R rows and C columns. This gives rise to RC categories. Each category contains N elements. (The quantities R , C , and N may be finite or infinite.) An urn sampling model is envisioned which samples r rows, c columns, and n elements from each of the rc selected categories. The expectations of the various mean squares in the analysis of variance table can then be calculated using indicator random variables. This generalized urn sampling model enables one to determine the appropriate error terms for testing row, columns, and interaction effects in the analysis of variance table.

4. Bayesian Inference

During the early 1960's Jerry gave increasing attention to the philosophy and methodology of Bayesian inference. In 1963 he co-authored a fundamental paper with Seymour Geisser comparing the fiducial and Bayesian distributions of the parameters associated with the multivariate normal distribution. The paper contained a large number of interesting results. Amongst these were that the posterior distribution of the population means can be made to coincide with either the fiducial or Hotelling T^2 distribution. However, these results can only be derived if different prior distributions are used. Another quite curious result is that there may exist confidence intervals which are individually, but not collectively, Bayesian. The most noteworthy example of this result is that both the confidence interval and region obtained from Student's and Hotelling's T^2 distributions, respectively, can both be derived from Bayesian considerations, but no common prior distribution exists which leads to both of them.

Jerry's views on Bayesian inference were very much influenced by De Finetti, Jeffreys, Lindley, and L. J. Savage. He published two landmark papers in the late 1960's (Cornfield, 1966; 1969). These papers attempted to synthesize Bayesian ideas with his own outlook which had been largely shaped by applications. His 1966 paper re-examined the sequential

conduct of clinical trials from the Bayesian point of view of Harold Jeffreys. In Jerry's outlook, a sequential clinical trial is not simply a test of a hypothesis, but a systematic method of observation, data reduction, and informal stopping rules. He considered the problem of testing a simple null hypothesis of the mean of a normal distribution versus a composite alternative in the context of sequential data collection. Adopting the ideas of Jeffreys, who makes a distinction between hypothesis testing and estimation, Jerry formulated the sequential problem by placing a concentration of prior probability at the simple hypothesis and a smooth prior for the alternate composite hypothesis.

He showed that with this formulation behavioral rules (stop or continue experimentation), based on repeated calculations of posterior probabilities, result in accepting the simple hypothesis when true with probability one as the sample size increases. Also, if the simple hypothesis is true, the chance of rejecting it can be made equal to an arbitrary probability level. This modification answered a criticism made by Peter Armitage that by virtue of the law of the iterated logarithm the use of the likelihood function will (as in the use of repeated tests of significance) result in rejecting any null hypothesis with probability unity. This paper also contained a number of practical results showing how the Bayesian formulation leads to closed sampling schemes and how sequential procedures can be developed to compare two binomial parameters.

Finally I wish to discuss Jerry's 1969 paper entitled "The Bayesian Outlook and Its Applications" which was published in *Biometrics*. Essentially this paper is a synthesis of his outlook on Bayesian inference. His philosophy is simply stated, "*Any inferential or decision process that does not follow from some likelihood function and some set of priors has objectively verifiable deficiencies.*" This is a paper which should be influencing the practice of statistics today. It is regarded by many as being the single most comprehensive paper advocating the use of Bayesian methods in applications. The paper synthesizes a large variety of ideas, taken from many investigators, into a connected whole.

Among the new ideas discussed were the use of relative betting odds (i.e., ratio of the posterior density of a point hypothesis to the density weighted by prior probabilities) and the connection between Bayesian inference and randomization. This last contribution is especially intriguing. It was first presented by Jerry at the International Statistical Institute meeting in Belgrade in 1965 and is published in the Proceedings of the meeting.

He considered the two population problems where there are $(m + n)$ experimental units, each unit being characterized by a control value (X) and an unknown treatment value ($Y = X + \theta$). It is possible to observe either the control or the treatment values on an experimental unit, but not both. The m units assigned to the control and n units assigned to the treatment are made by randomization. What inferences can be made on $\theta = Y - X$? This is essentially the setting for use of the permutation test to compare two populations. Jerry derived the likelihood of θ and showed that it is constant (and thus uninformative for making inferences on θ). However, by assuming appropriate prior probabilities, the posterior distribution of θ is shown to be modulated by the likelihood. Thus we have a situation where the likelihood is uninformative and is only of use if posterior probabilities are calculated.

5. Concluding Remarks

In this all-too-brief paper, I have attempted to describe the breadth of Jerry's intellectual contributions to statistical theory. It is most appropriate to end this review by quoting from Jerry's American Statistical Association Presidential Address.

"I must admit that in my more Walter Mitty-like moments I have contemplated the cancer cures I might have discovered had I gone into one of the biomedical sciences, and

similar marvels for other fields. But advancing years bring realism, and I am convinced that the outcome of the random walk which made me a statistician was inevitable. The only state that was an absorbing one was statistics ...”.

ACKNOWLEDGEMENT

This paper is supported in part by grants CA-10810 and CA-23415 from the National Cancer Institute.

REFERENCES

- Cornfield, J. (1944). On samples from finite populations. *Journal of the American Statistical Association* **37**, 236–239.
- Cornfield, J. (1965). A note on the likelihood function generated by randomization over a finite set. *Proceedings of the 35th Session of the International Statistical Institute, Belgrade, Yugoslavia*.
- Cornfield, J. (1966). A Bayesian test of some classical hypotheses with applications to sequential clinical trials. *Journal of the American Statistical Association* **61**, 577–594.
- Cornfield, J. (1969). The Bayesian outlook and its applications. *Biometrics* **25**, 617–657.
- Cornfield, J. (1975). A statistician's apology. *Journal of the American Statistical Association* **70**, 7–14.
- Cornfield, J. and Tukey, J. W. (1956). Average values of mean squares in factorials. *Annals of Mathematical Statistics* **27**, 907–949.
- Dantzig, G. B. and Laderman, J. (1947). Rand Corporation unpublished report.
- Dorfman, R., Samuelson, P. A., and Solow, R. M. (1958). *Linear Programming and Economic Analysis*, New York: McGraw Hill.
- Eisenhart, C. (1947). The assumptions underlying the analysis of variance. *Biometrics* **3**, 1–21.
- Geisser, S. and Cornfield, J. (1963). Posterior distributions for multivariate normal parameters. *Journal of the Royal Statistical Society, Series B* **25**, 368–376.
- Pitman, E. J. G. (1938). Significance tests which may be applied to samples from any population, III. The analysis of variance test. *Biometrika* **29**, 322–335.
- Stigler, G. J. (1945). The cost of subsistence. *Journal of Farm Economics* **27**, 303–314.
- Tukey, J. W. (1949). Interaction in a row-by-column design. *Memorandum Report 18, Statistical Research Group*, Princeton University, Princeton, New Jersey.
- Welch, B. L. (1937). On the Z-test in randomized blocks and latin squares. *Biometrika* **29**, 21–52.
- Wilk, M. B. and Kempthorne, O. (1955). Fixed, mixed, and random models. *Journal of the American Statistical Association* **50**, 1144–1167.